Exercise 1

1. SELECT *
FROM employees;

Id	First_name	last_name	Department	Salary	hire_date	City
1	Alice	Green	IT	70000 45000	2020-01-10	Johannesburg
2	Brian	Lee	HR	65000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. SELECT DISTINCT department
FROM employees;

Department
IT
HR
Finance
Marketing

3. SELECT first_name,
last_name
FROM employees
ORDER BY salary DESC;

First_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. SELECT id,
first_name,
last_name, salar
ORDER BY SALARY DESC
LIMIT 3;

Id	First_name	Last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. SELECT id,
first_name,
last_name, department
FROM employees
WHERE department='IT';

Id	First_name	Last_name	Department
1.	Alice	Green	IT
5	Eva	Naidoo	IT

7. 6. SELECT id,
first_name,
last_name
FROM employees
WHERE department='~~IT~~' HR' OR
department='Marketing';

Id	First_name	last_name	Department
2	Brian	Lee	HR
4	David	Mokoena	marketing

no 7. OUTPUT

Id	First_name	Last_name	Department
1.	Alice	Green	IT
4	David	Mokoena	Marketing
5	Eva	Naidoo	IT

6. SELECT Id,
first_name,
Last_name
FROM employees
WHERE department='Finance'
AND Salary > 60000;

Id	First_name	Last_name	Department	Salary
3.	Cathy	Zulu	Finance	65 000

8. SELECT Id,
first_name,
last_name
FROM employees
WHERE department NOT ='IT';

Id	firstname	last_name	Department
2.	Brian	Lee	HR
3	Cathy	Zulu	~~Marketing~~ Finance
4	David	Mokoena	Marketing

9. SELECT Id,
first_name,
last_name
FROM employees
WHERE department IN('IT, HR, Finance');

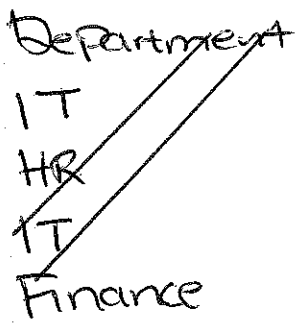
Id	First_name	last_name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT Id,
first_name,
last_name
FROM employees
WHERE department='IT'
AND Salary 76500
AND City = 'Johannesburg';

Id	First_name	last_name	Department	Salary	City	
1.	Alice	Green	IT	70000	Johannesburg	
5	Eva	Naidoo	IT	72000	Johannesburg	

Exercise 2

1. SELECT DISTINCT department
FROM Students;



Department
IT
HR
Finance

2. SELECT department,
AVG(age) AS Avg_age
FROM Students
GROUP BY department;

Department	Avg_age
IT	20,5 (20+21=41/2)
HR	22,0 (22+22=44/2)
Finance	23,0 (23/1)

3. SELECT department,
COUNT(*) AS student_count
FROM Students
WHERE COUNT(*) > 1
~~Checking~~

Department	Student_count
IT	2
HR	2

4. SELECT Student_Id,
name,
age,
department
FROM Students
WHERE age BETWEEN 21 AND 23;

Student id	name	age	Department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student_id,
name,
age,
department
FROM Students
WHERE department IN('HR','IT')
AND age 721

Student_Id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department,
SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

Department	total_credits
IT	11

7. SELECT course_id,
course_name,
department,
credits
FROM courses
WHERE credits < 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id,
course_name,
Credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9.
```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84,6

10.
```
SELECT course_id,
       COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. SELECT department,
Sum(salary) AS total_salary,
Sum(bonus) AS total_bonus
FROM salaries
GROUP BY departments;

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department
AVG(Salary) AS avg_salary
FROM salaries
GROUP BY departments
HAVING AVG(Salary) 75000;

department	avg_salary
IT	61000
Finance	70000

60000 + 62000 = 122000/2 = 61000 > IT

Finance = 70000/1 = 70000

13. SELECT employee id,
name,
Salary,
bonus,
Salary + bonus AS total_compensation
FROM Salaries
WHERE Salary + bonus 76000;

employee_id	name	Salary	bonus Salary	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. SELECT department,
SUM(budget) AS total_budget
AVG(budget) AS avg_budget
FROM Projects
GROUP BY department
HAVING AVG(budget) > 70000;

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

15. SELECT project_id,
project_name,
department,
budget
FROM Projects
WHERE budget BETWEEN 5000 AND 120000
AND department <> 'Marketing';

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000

EXERCISE 3

1.
```
SELECT product_name,
       price,

    CASE
        WHEN price > 1000 THEN 'Expensive'
        WHEN price BETWEEN 100 AND 1000
         THEN 'Mid_range'
    ELSE 'Budget'
END AS ~~product~~ price_category
FROM products;
```

~~product~~ name	price	price_category
Laptop	12000	Expensive
Phone	800	Mid_range
Keyboard	45	Budget
Monitor	300	Mid_range
Mouse	25	Budget

2.
```
SELECT customer_name,
       amount,

    CASE
        WHEN amount >= 1000 THEN 'High Value'
        WHEN amount BETWEEN 500 AND 999.99
         THEN 'Medium Value'
    ELSE 'Low Value'
END AS order_value_category
FROM orders;
```

customer_name	amount	order_value_category
Alice	150	Low Value
Bob	560	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	12000	High Value

3. SELECT emp_name,
department,
Salary,

CASE
WHEN department='IT' and Salary >80000
THEN 'Senior IT'
WHEN department='HR' and Salary ~~85000~~ 5500
THEN 'Experienced HR'
ELSE 'staff'
END AS position level
FROM employees;

EMPname	department	salary	position_level
1. John	IT	85000	~~Senior IT~~
2. Lucy	~~finance~~	95000	Senior IT
3. Mark	IT	75000	Experienced H

Emp_name	department	Sarey	position_level
John	IT	8500	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	STAFF
Tom	HR	55000	STAFF

14.
```
SELECT student_name,
       Score,
    CASE
      WHEN Score >= 90 THEN 'A'
      WHEN Score >= 80 THEN 'B'
      WHEN Score >= 70 THEN 'C'
      WHEN Score >= 60 THEN 'D'
    ELSE 'F'
  END AS grade
FROM students;
```

student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. SELECT delivery_id,
delivery_time_minutes,
CASE
WHEN delivery_time_minutes <=30 THEN 'Fas
WHEN delivery_time_minutes <= 60 THEN
'ON Time'
WHEN ELSE 'Late'
END AS performance
FROM deliveries;

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	30	Late
5	100	Late

6. SELECT issue_type,
priority,
CASE
WHEN 3 THEN 'High'
WHEN 2 THEN 'medium'
WHEN 1 THEN 'Low'
END AS priority_label
FROM tickets;

Issue_type	priority	Priority_label
Login Issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7.
```
SELECT student_id,
       (days_present * 100 / total_days) AS
       attendance_percentage,
    CASE
        WHEN (days_present * 100.0 / total_days)
        >= 90 THEN 'Excellent'
       WHEN (days_present * 100.0 / total_days)
       >= 75 THEN 'Good'
    ELSE 'Needs Improvement'
    END AS attendance_status
FROM attendance
```

Student_id	attendance_percentage	attendance status
1	90.0	Excellent
2	60.0	Needs Improvement
3	96.0	Excellent
4	50.0	Needs Improvement
5	100.0	Excellent

8. SELECT product_id,
stock_qty,

CASE
WHEN stock_qty = 0 THEN 'Out of stock'
WHEN stock_qty BETWEEN 1 AND 5
THEN 'Low Stock'
ELSE 'In Stock'
FROM END AS Stock Status
FROM products_inventory;

product_id	stock_qty	stock_status
1	5	Low stock
2	0	Out of stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

9. SELECT subject,
enrolled_students,
CASE
WHEN enrolled_students >= 25 THEN 'Large'
WHEN enrolled_students >= 10 THEN 'Medium'
ELSE 'Small'
END AS class_size_category
FROM classes;

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10.
```
SELECT payment_id,
       payment_method,
       amount,
    CASE
       WHEN payment_method = 'Cash' AND
       amount >= THEN 'Eligible for Discount'
  ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount

DATE FUNCTIONS: EXERCISE 4

1. SELECT order_id,
customer_id,
order_date,
DAYNAME(order_date) AS day_name
FROM orders;

order_id	customer_id	order_date	day_name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

2. SELECT customer_id,
customer_name,
signup_date,
MONTHNAME(signup_date) AS signup_month_name
FROM customer_signups;

customer_id	customer_name	signup_date	signup_month_name
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

3. SELECT sale_id,
product_name,
Sale_date,
MONTH (Sale_date) AS Sale_month
FROM Sales;

Sale_id	product_name	Sale_date	Sale_month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

4. SELECT transaction_id,
Customer_id,
transaction_date,
YEAR (transaction_date) AS transaction_year
FROM transactions;

transaction_id	Customer_id	transaction_date	transaction_year
T001	101	2024-12-15	2024
T002	102	2025-01-10	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

5. SELECT delivery_id,
Customer_id,
delivery_date,
DAY (delivery_date) AS day_of_month
FROM deliveries;

delivery_id	customer_id	delivery_date	day_of_month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

6. SELECT employee_id,
employee_name,
department,
CURRENT_DATE() AS today_date
FROM employees;

employee_id	employee_name	department	today_date
301	Nandi	Sales	2026-07-19
302	Bran	IT	2026-07-19
303	Lerato	Finance	2026-07-19
304	Sipho	HR	2026-07-19
305	Aisha	Marketing	2026-07-19

7. SELECT order_id,
customer_id,
order_date_text,
TO_DATE(order_date_text, 'YYYY-MM-DD') AS
order_date
FROM online_orders;

order_id	customer_id	order_date_text	order_date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-04-10

8. SELECT payment_id,
customer_id,
payment_date,
TO_CHAR(payment_date, 'YYYY-MM-DD') AS
formatted_payment_date
FROM payment_dates;

Payment_id	customer_id	payment_date	Formatted_payment_date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

9. SELECT customer_id,
customer_name,
last_purchase_date,
DATEDIFF(CURRENT_DATE(), last_purchase_date) AS
days_since_last_purchase
FROM customer_purchases;

Customer_id	customer_name	last_purchase_date	days_since_last_purchase
501	John	2026-05-01	79
502	Mary	2026-05-10	70
503	Peter	2026-05-15	65
504	Sarah	2026-05-20	60
505	Thabo	2026-05-25	55

EXERCISE 5

1. SELECT employee_id,
name,
IFNULL(salary, 0) AS salary_wit
default
FROM employees;

employee_id	name	Salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2. SELECT employee_id,
name,
IFNULL(department, 'Not Assigned') AS
department_name
FROM employees;

employees_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

3. SELECT order_id,
Customer_id
FROM orders
WHERE ISNULL(customer_id);

order_id	Customer_id
103	NULL

4. SELECT order_id,
Customer_id
IFNULL(CAST(delivery_date AS CHAR),
'Pending') AS delivery_status
FROM orders;

order_id	Customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

5. ~~SET~~ SELECT student_id,
name,
IFNULL(grade, 0) AS final_grade
FROM Students;

OUTPUT

Student_id	name	final_grade
1	Ethan	55
2	Maya	0
3	Olivia	90

6. SELECT COUNT(*) AS not_graded_count
FROM students
WHERE grade IS NULL;

OUTPUT

not_graded_count
1

7. SELECT product_id,
name,
price - IFNULL(discount, 0) AS final_price
FROM products;

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

8. SELECT COUNT(*) AS missing_email_count
FROM customers
WHERE email IS NULL;

missing_email_count
2

9. SELECT customer_id,
name,
IFNULL(email, 'NO Email') AS email_display
FROM customers

customer_id	name	email_display
1	Linda	No email
2	Joseph	joseph@mail.com
3	Mia	No email

10. SELECT payment_id,
IFNULL(method, 'Unknown') AS method_display,
status
FROM payments;

payment_id	method_display	status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed

11. SELECT item_id,
item_name
IFNULL(quantity, 0) AS quantity_checked
FROM inventory;

12.

item_id	item_name	quantity_checked
1.	Pen	0
2.	Notebook	150
3.	Eraser	0

12. SELECT emp_id,
COALESCE(bonus, commision) AS
first_available_reward
FROM employees_extra;

emp_id	first_available_reward
1	300
2	100
3	NULL

13. SELECT COUNT(*) AS no_room_count
FROM classes
WHERE room IS NULL;

no_room_count
2

14. SELECT student_id,
date,
IFNULL(status, 'Not Marked') AS attendance_
status
FROM attendance;

student_id	date	attendance_status
1	2025_04_01	Not marked
2	2025_04_01	Present
3	2025_04_01	Absent

15. SELECT account_id,
IFNULL(account_type, 'Unknown') AS
type_display,
IFNULL(balance, 0) AS balance_checked
FROM bank_accounts;

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

16. SELECT project_id,
title,
IFNULL(CAST(start_date AS CHAR), 'TBD')
AS start_display
FROM projects;

Project_id	title	start_display
1	website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

17. SELECT review_id,
product_id
IFNULL(comment, 'NO Comment') AS
comment_display,
IFNULL (rating, 0) As rating_display
FROM reviews;

review_id	product_id	comment_display	rating_display
1	501	Great Product	4
2	502	No comment	0
3	503	Works fine	3

18. SELECT supplier_id,
name,
COALESCE(phone, alt_phone, 'NO Contact')
As contact_number
FROM suppliers;

Supplier_id	name	Contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	NO Contact

19. SELECT user_id,
IFNULL(theme, 'Light') AS theme_set,
IFNULL(language 'English') AS language_set,
IFNULL(timezone, UTC') AS timezone_set
FROM usersettingss;

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

20. SELECT record_id,
machine_id,
IFNULL(issue, 'Unkown Issue') AS issue_log,
IFNULL(technician, 'Not Assigned') AS
technician_togname
FROM maintarance;

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex

Exercise 3 continues

10.
```
SELECT order_id,
       customer_id,
       order_date,
       DATEADD(DAY, 7, order_date) AS expected_delivery_date
FROM Shipping_orders;
```

order_id	customer_id	order_date	expected_delivery_date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

11.
```
SELECT booking_id, customer_id, booking_date,
       YEAR(booking_date) AS booking_year,
       MONTH(booking_date) AS booking_month,
       DAY(booking_date) AS booking_day
FROM bookings;
```

booking_id	customer_id	booking_date	booking_year	booking_month	booking_day
B001	101	2026-01-12	2026	1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	9
B005	105	2026-05-27	2026	5	27

12. SELECT order_id, customer_id, order_date,
YEAR(order_date) AS order_year, amount
FROM yearly_orders
WHERE YEAR(order_date) = 2026;

order_id	customer_id	order_date	order_year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

13. SELECT order_id, customer_id, order_date,
MONTH(order_date) AS order_month, amount
FROM monthly_orders
WHERE MONTH(orders_date) = 3;

order_id	customer_id	order_date	order_month	amount
8003	103	2026-03-10	3	800
8004	104	2026-03-25	3	1500

14. SELECT subscription_id, Customer_id, Start_date,
LAST_DAY(Start_date) AS month_end_date
FROM Subscriptions;

Subscription_id	customer_id	start_date	month_end_date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

15. SELECT send_id, customer_id, send_date,
DATE_TRUNC('month', send_date) AS
month_start_date
FROM campaign_sends;

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

16. SELECT invoice_id, customer_id, invoice_date,
TO_CHAR(invoice_date, 'Month YYYY') AS
invoice_month_year
FROM invoice_dates;

invoice_id	customer_id	invoice_date	invoice_month_year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

17. SELECT customer_id, customer_name, date_of_birth
FLOOR(DATEDIFF(CURRENT_DATE(), date_of_birth
/365.25) AS customer_age
FROM customer_birthdays;

customer_id	customer_name	date_of_birth	customer_age
901	John	1998-05-10	28
902	Mary	1990-08-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Thabo	2000-07-30	25

18
```
SELECT order_id, customer_id, order_date,
       DAYNAME(order_date) AS day_name,
    CASE
      WHEN DAYNAME(order_date) IN
        ('Saturday', 'Sunday') THEN 'Weekend'
      ELSE 'Weekday'
    END AS day_type
FROM weekend_orders
```

order_id	customer_id	order_date	day_name	day_type
9001	101	2026-05-01	Friday	Weekday
9002	102	2026-05-02	Saturday	Weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	Monday	Weekday
9005	105	2026-05-05	Tuesday	Weekday

19.
```
SELECT transaction_id, customer_id, transaction_date,
       QUARTER(transaction_date) AS transaction_
       quarter, amount
FROM quarterly_transactions;
```

transaction_id	customer_id	transaction_date	transaction_quarter	amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-10	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

20. SELECT order_id, customer_id, order_date,
DATEDIFF(CURRENT_DATE(), order_date)
AS days_since_order, amount
FROM recent_orders
WHERE DATEDIFF(CURRENT_DATE(), order_date) > 730;

order_id	customer_id	order_date	days_since_order	amount
R001	101	2026-04-01	109	500
R002	102	2026-04-15	95	1200
R003	103	2026-05-01	79	800
R004	104	2026-05-10	70	1500
R005	105	2026-05-25	55	2000

Bonus Question

SELECT customer_id, customer_name, last_purchase_date,
DATEDIFF(CURRENT_DATE(), last_purchase_date)
AS days_since_last_purchase,
CASE
WHEN DATEIFF(CURRENT_DATE(),
last_purchase_date) <= 30
THEN 'Active Customer'

(vi)

```
WHEN DATEDIFF(CURRENT-DATE(),
last_purchase_date) <= 90
THEN 'At Risk Customer'
ELSE 'Inactive Customer'
END AS Customer_status
FROM customer_recency
```

Current_id	Customer_name	last_purchase_date	days_since_last_purchase	Custom Stat
1001	John	2026-05-25	55	At Risk C
1002	Mary	2026-05-10	70	At Risk Cu
1003	Peter	2026-04-01	109	Inactive Cust
1004	Sarah	2026-02-15	154	Inactive Custo
1005	Trabo	2025-12-20	211	Inactive Cust